

Real Time Systems Laboratory

ELCT 302-001

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Lab Notebook

Group 2 (Short Circuits)

Driving Team

Department of Electrical Engineering

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Table of Contents

Table of Contents	2
Lab 01: Open-Loop Control & Obtaining Feedback	3
General Objectives	3
Main Specifications	3
Tasks: Open-Loop Control	3
Tasks: Speed Measurement	3
Background	4
Calculations	10
Code	10
Results	12
Problems Encountered	17
Questions	18
Time sheet	18
Lab 02: Model Identification	20
General Objectives	20
Main Specifications	20
Tasks: Part I	20
Tasks: Part II	20
Background	21
Component Specifications (Data Sheets)	22
Mbed Pinout	22
Mbed Datasheet	22
MOSFET (IRF540NPBF) Datasheet	22
Schottky Diode (MBR1060-E3/45) Datasheet	22
Motor (RN260-CN-18130) Datasheet	22
Voltage Regulator (L7805CV) Datasheet	23
Low-Side Gate Driver (TC4428EPA) Datasheet	23
Optical Sensor (OPB829AZ) Datasheet	23
Frequency to Voltage Converter (LM2907N/NOPB) Datasheet	23

Lab 01: Open-Loop Control & Obtaining Feedback

Motor Control (Group A)

General Objectives

- Use programming techniques to solidify understanding of microcontroller features.
 - API is available on Mbed.com for the KL25Z GPIO drivers under MbedOS section.
- Demonstrate understanding and implementation of open-loop DC motor control
 - Single switch buck converter.
- Design and implement a circuit to create a feedback signal for motor speed
 - Design and implement an encoder.
 - Design a frequency to voltage converter.
- Determine speed to voltage relationship of the DC motor.

Main Specifications

- Motor voltage: 7.2V
- Switching frequency: ≥ 20 kHz
- Speed range: 0-100% (Design to match the DC motor's rated voltage!!!)
- External voltage (speed) reference: 0-3.3V
- Feedback: Incremental Encoder

Tasks: Open-Loop Control

- Research data sheets for components and understand why they are used (their ratings)
- Design an adjustable voltage divider (Potentiometer) circuit with an output ranging from 0 - 3.3V.
- Write code that reads in the voltage divider output as an ADC number and writes the ratio to the desired maximum as the PWM duty cycle.
- Upload and verify with microcontroller. (Keep in mind the DC motor's voltage rating!!) (Hint: Lab 0)
- Build buck converter and verify its operation

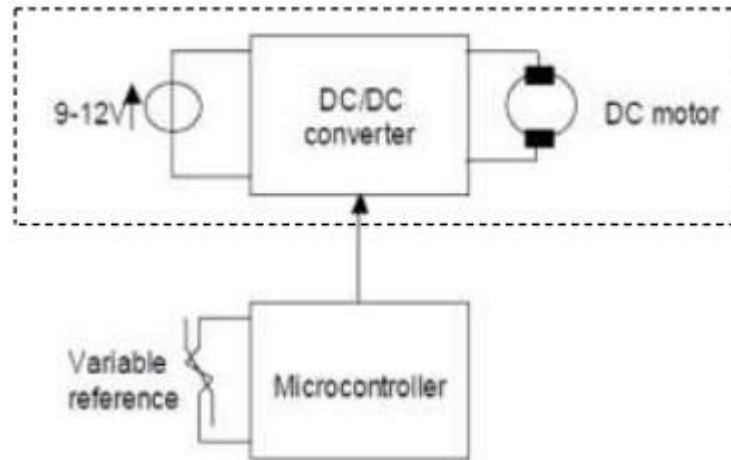
Tasks: Speed Measurement

- Research the LM2907 and Optical Switch sensor datasheets; understand the operation of both circuits
- Design frequency-to-voltage converter (LM2907) and circuitry for Optical Switch sensor using information from lecture and datasheets
- Tabularize motor control and feedback measurement range and summarize results:
 - Vref (V), reference voltage measured from potentiometer
 - Vref ADC, reference voltage digital conversion
 - Duty Cycle (0-1), positive duty cycle
 - Fencoder, Encoder frequency (Hz)
 - Vout (V), speed measurement in volts from speed sensor
 - Vout ADC, speed measurement voltage digital conversion

- Wheel Speed (rad/s) o Motor Speed (rad/s)
- Average motor voltage (V): $V_{dc} \cdot \text{duty-cycle}$ (0-1)

Background

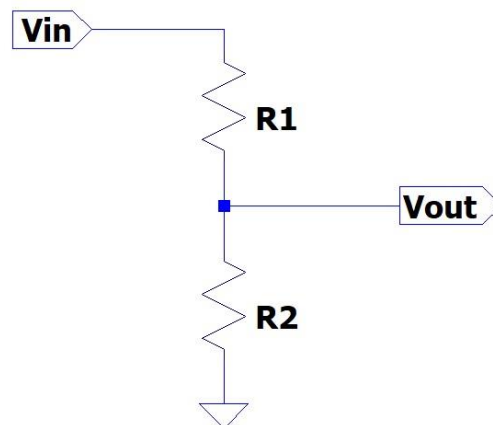
- **Circuit Diagram**



- **Reference Voltage**

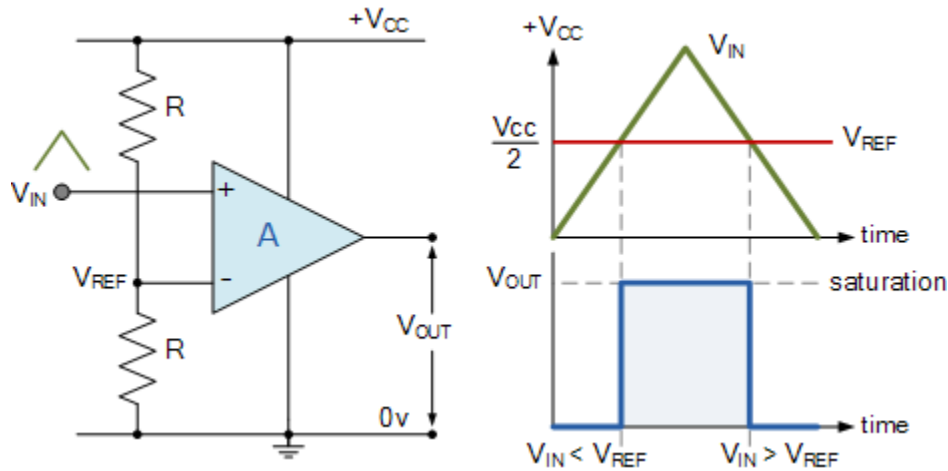
For the external reference voltage, a potentiometer is used as a voltage divider. The input to the potentiometer will be from the 3.3V pin of the KL25Z microcontroller. The output (voltage dependent on the potentiometer wiper location) will be directed into an analog input pin of the microcontroller.

$$V_{out} = \frac{R_2}{R_1 + R_2} \cdot V_{in}$$



- **Microcontroller**
 - Analog-Digital-Converter (ADC)

A basic ADC circuit is comprised of comparators. A voltage input (V_{IN}) signal is applied to the input of the comparator, while a reference voltage (V_{REF}) is applied to the other. A comparison of the voltages is made, and the output of the comparator will either be high or low.



One comparator could be considered a 1-bit ADC. To increase the bit size of the ADC, comparators are stacked. The general equation is:

$$n - \text{bit ADC} = 2^n - 1$$

The KL25Z utilizes a 16-bit (65,536 distinguishable outputs) ADC and uses an *AnalogIn* API to read an external voltage applied to an analog input pin (3.6V max input voltage to pin). The resolution of the ADC can be calculated to find the amount of voltage required to make a change in the digital output.

$$\frac{V_{max}}{\text{distinguishable outputs}}$$

The voltage is read as a fraction of the system voltage. This value is a floating point from 0.0(VSS) to 1.0(VCC). A change in analog input voltage, given by the equation above, will create a change in the digital output voltage by;

$$\frac{1}{\text{distinguishable outputs}}$$

○ Pulse Width Modulation (PWM)

PWM is a train of continuous pulses, where the start of each pulse is separated by a constant time. The time it takes for each pulse is defined as the period (T) of the waveform and is inversely proportional to the frequency (f).

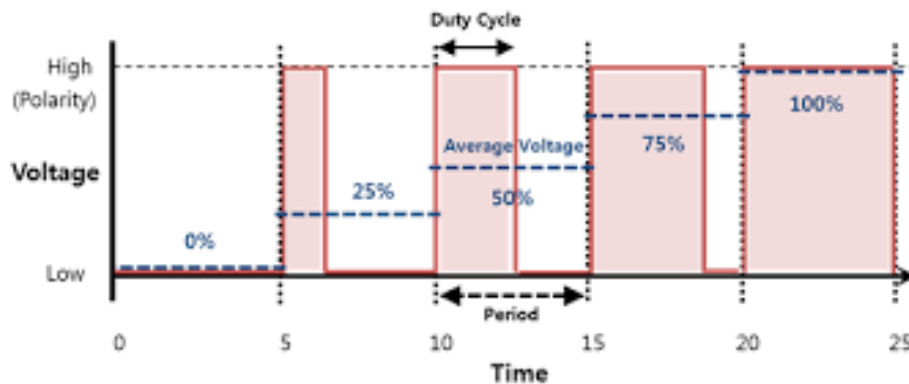
$$T = \frac{1}{f}$$

The duty cycle is the relationship between the time on and time off

$$D = \frac{T_{on}}{T_{on} + T_{off}} = \frac{T_{on}}{T}$$

With the output waveform of the PWM being a square wave, the voltage output of the PWM will be equal to the average voltage.

$$V_{out} = V_{in} \cdot D$$



It should be noted that the KL25Z microcontroller's 16-bit ADC can operate at frequencies of 2 to 12 MHz or up to 83.33 ns. While the GPIO can operate at up to 62.5 MHz or 16 ns.

- **DC/DC Converter**

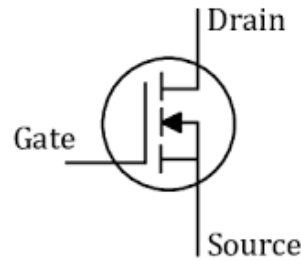
To generate and control voltage for the DC motor a DC/DC converter will be utilized. The motor's speed is linearly proportional to the voltage, so the voltage needs to be as accurate as possible. Using the PWM signal from the microcontroller and a switch, a desired voltage can be created to control the motor speed. The desired DC signal needs to filter out the lower fundamental frequencies and harmonics.

- **Low-Side Gate Driver**

One inverting, one non-inverting = flipped waveform.

- **MOSFET**

An N-channel MOSFET will be utilized as the switch for the converter. A MOSFET has a gate, source, drain, and body. Typically, the source and body are internally connected. When a voltage is applied to the gate-source (V_{GS}), and it is higher than the threshold voltage (V_{th}) of the MOSFET, current will flow.



Drain current (i_D) is dependent on the K value and voltage threshold of the MOSFET and the voltages applied at the gate and drain. It is also dependent on the operating region as well.

$$K = \frac{W}{L} \cdot \frac{KP}{2}$$

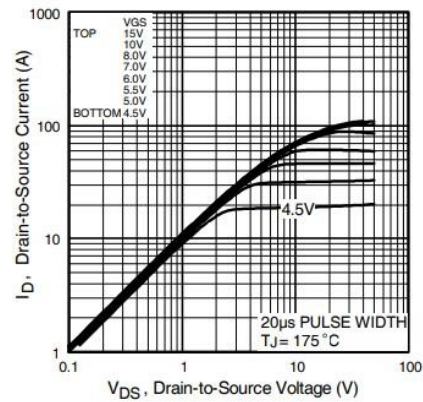
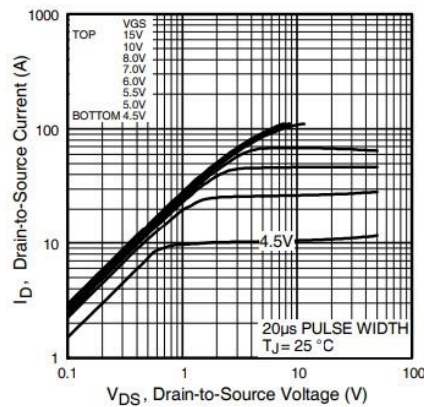
$$KP = \mu_n \cdot C_{ox}$$

There are three regions the MOSFET operates. The cut-off region occurs when $V_{GS} < V_{th}$. The triode region occurs when, $V_{GS} > V_{th}$ but the drain-source voltage (V_{DS}) is less than the difference of the two $V_{DS} < (V_{GS} - V_{th})$. The current in the triode region is,

$$i_D = K[2(V_{GS} - V_{th})V_{DS} - V_{DS}^2]$$

The saturation region occurs when $V_{DS} \geq (V_{GS} - V_{th})$. The current in the saturation region is,

$$i_D = K(V_{GS} - V_{th})^2$$



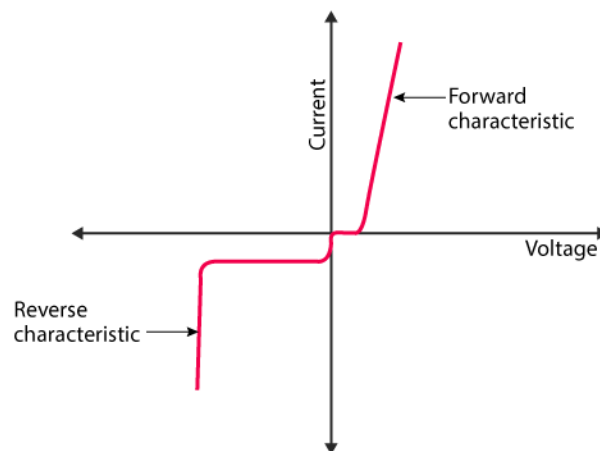
It is important to note that the specifications for the MOSFET utilized has a pulse width $\leq 400\mu s$; duty cycle $\leq 2\%$.

○ Schottky Diode

The Schottky diode is a type of metal–semiconductor junction diode. This diode is formed by the junction of a semiconductor with a metal. Schottky diode offers fast switching action and has a low forward voltage drop.

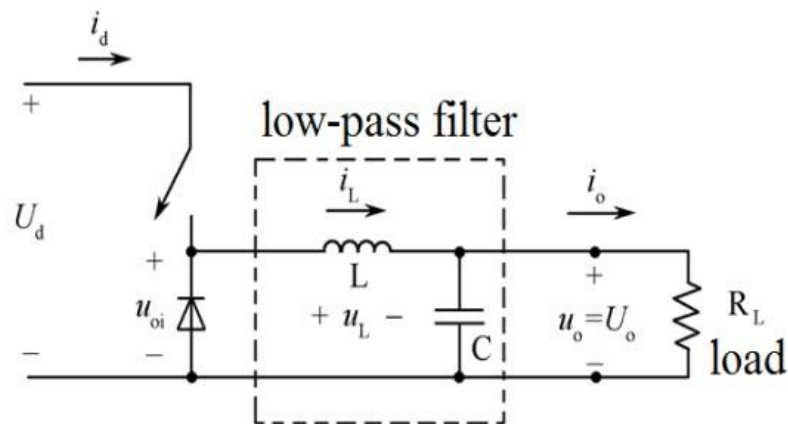


Current is the dependent variable while voltage is the independent variable in the Schottky diode.



○ Buck Converter

The motor utilized operates at a lower voltage than the voltage source. The buck converter will step down the voltage supplied to the operating voltage of the motor.

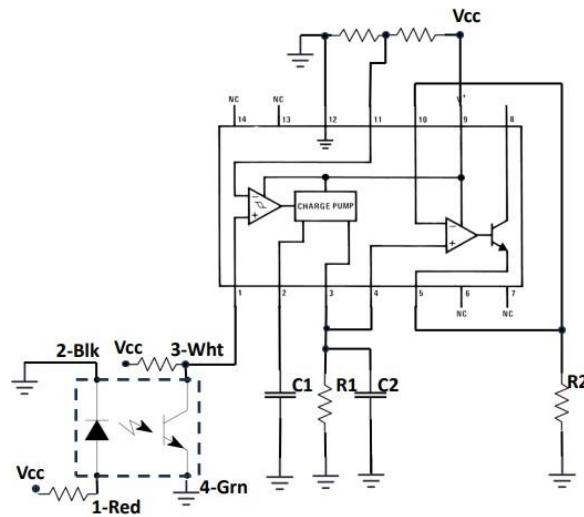


To reduce voltage ripple, filters made of capacitors are normally added. However, the motor's own inductance acts like a filter, storing energy during the “on” cycle while releasing it at a rate corresponding to the input or reference signal. The buck converter's switch can be located on the high-side or the low-side. The low-side configuration will be used with the MOSFET discussed earlier.

○ Speed Measurement

To calculate the speed of the vehicle, the difference between the desired speed and real speed must be measured. An optical sensor will determine the frequency of tire rotation. Phototransistor switching occurs when an opaque object passes through the slot of the device. The frequency of the output depends on the speed between on and off transitions.

This frequency will need to be changed into a voltage, so that the “actual” speed voltage can be compared to the “desired” speed voltage. A frequency to voltage converter can accomplish this task. The input frequency provided by the sensor is filtered through a Schmitt trigger. This reduces the noise from the input signal. The filtered signal is then sent to a charge pump that converts the signal frequency to a DC voltage.



Additional components are added to the frequency to voltage converter. Resistors and capacitors are needed to supply the charge pump with internal compensation. The values of the components can alter the response time, linearity, and ripple of the output. The following calculations are used to:

$$V_o = V_{cc} \cdot f_{in} \cdot C_1 \cdot R_1 \cdot K \quad (\text{Where } K = 1)$$

$$R_1 \geq \frac{V_{o,max}}{i_{3,min}}$$

$$f_{max} = \frac{I_2}{C_1 \cdot V_{cc}}$$

$$V_{ripple} = \left(\frac{V_{cc}}{2}\right) \cdot \left(\frac{C_1}{C_2}\right) \cdot \left(1 - \left(\frac{V_{cc} \cdot f_{in} \cdot C_1}{I_2}\right)\right)_{pk-pk}$$

$$= \left(\frac{V_{cc}}{2}\right) \cdot \left(\frac{C_1}{C_2}\right) \cdot \left(1 - \left(\frac{V_o}{I_2 \cdot R_1}\right)\right)_{pk-pk}$$

Calculations

- ADC

$$\frac{3.3V}{65,536} = 50\mu V$$

$$\frac{1}{65,536} = 15\mu V$$

- Tachometer

Resistors for sensor

$$R_1 \geq \frac{3.3V}{180\mu A} = 18.33k\Omega$$

$$C_1 \geq 500pF$$

$$R_2 = 10k\Omega$$

$$C_2 = 1\mu F$$

$$V_o = 5V \cdot 1nF \cdot 20k\Omega \cdot 1 \cdot f_{in}$$

Code

```
#include "mbed.h"
```

```
//#include "unistd.h"
```

```
//#include "pinmap.h"
```

```
// Voltage into the microcontroller from the voltage divider
```

```
AnalogIn potV(A0);
```

```
// Voltage into the microcontroller from the right motor speed
```

```
AnalogIn rmV(A1);
```

```
// Voltage into the microcontroller from the right motor speed
```

```

AnalogIn lmV(A2);

// Pulse width to the right motor
PwmOut poutRM(D0);

// Pulse width to the right motor
PwmOut poutLM(D1);


// Instantiate variables
float vref = 0.0;
float vrm = 0.0;
float vlm = 0.0;


// Code for print function on tera term
static BufferedSerial serial_port(USBTX, USBRX, 9600);
FileHandle *mbed::mbed_override_console(int fd)
{
    return &serial_port;
}


// Main code
int main() {

    // Set period of pulse widths
    //poutRM.period_us(40);
    //poutLM.period_us(40);
    poutLM.period(1.0f/25E3f);
    poutRM.period(1.0f/25E3f);


    // While motor running
    while (1) {

```

```

// Read the incoming voltage from pot
vref = potV.read();

// Print digital voltage input
printf("Digital voltage input %1.4f\r", vref);

//Send pulse width according to inverted duty cycle. The waveform is flipped at the gate driver.
poutRM.write(1.0f-vref);
poutLM.write(1.0f-vref);

// Read incoming voltage from wheel speed
vrm = rmV.read();
vlm = lmV.read();

// Print wheel speed voltage values
printf("Voltage right motor %1.4f\r", vrm);

printf("Voltage left motor %1.4f\r", vlm);

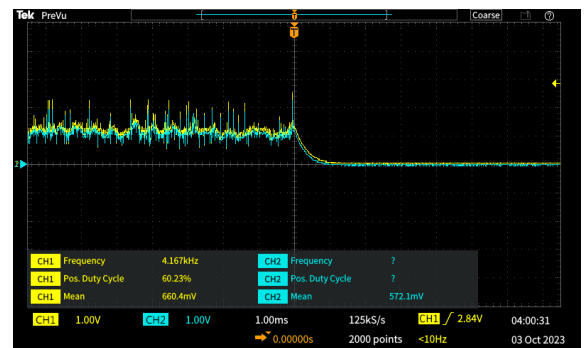
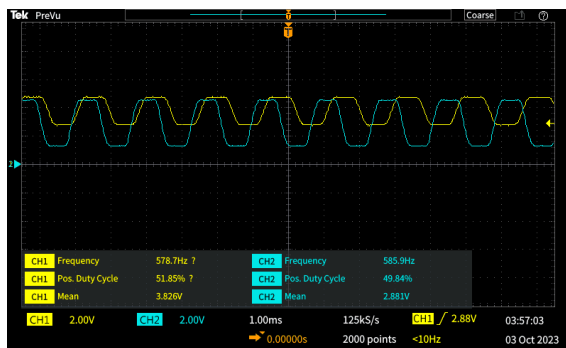
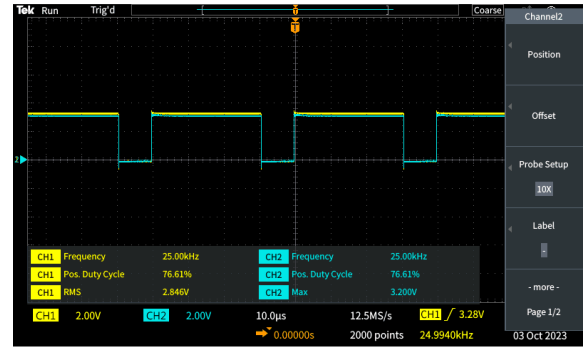
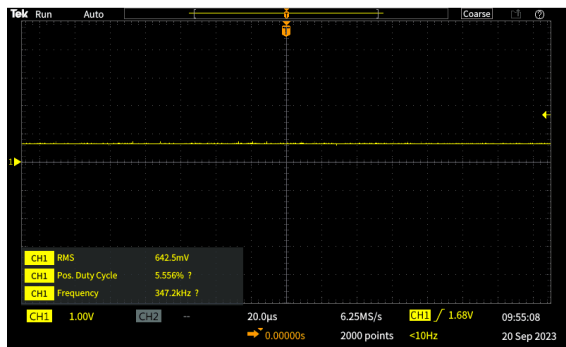
//Wait
//usleep(40);

}
}

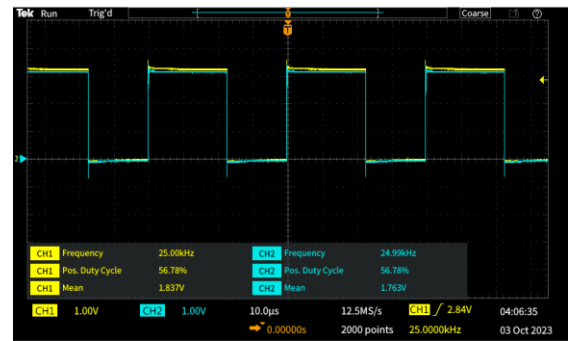
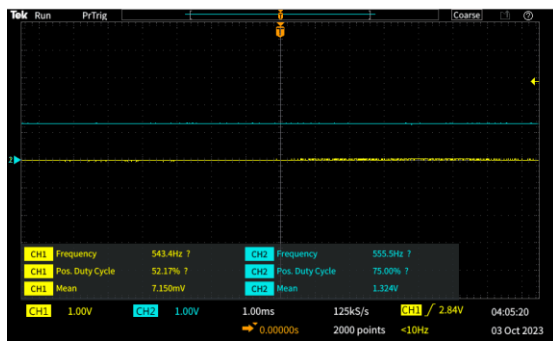
```

Results

- $0.6 V_{ref}$

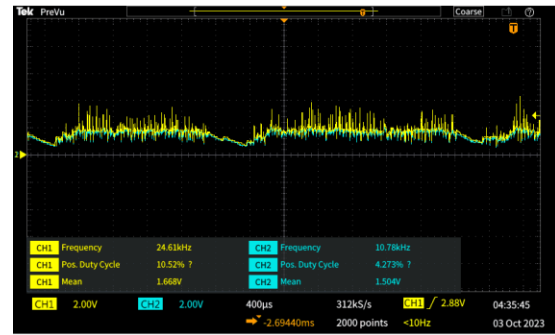


- $1.32 V_{ref}$



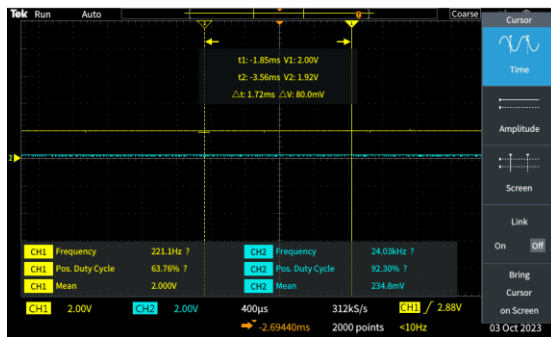


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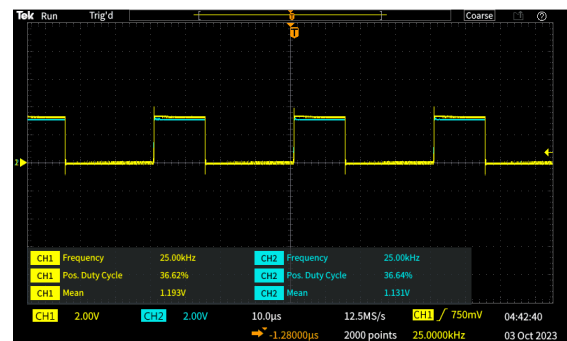


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- $2.0 V_{ref}$



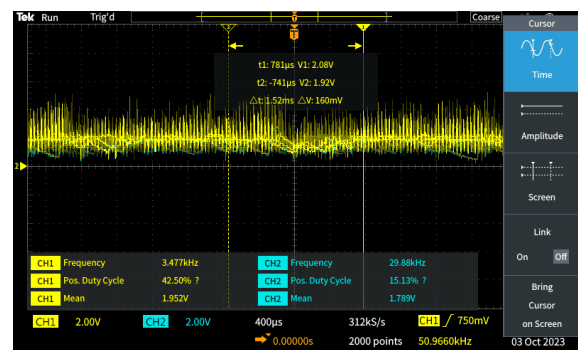
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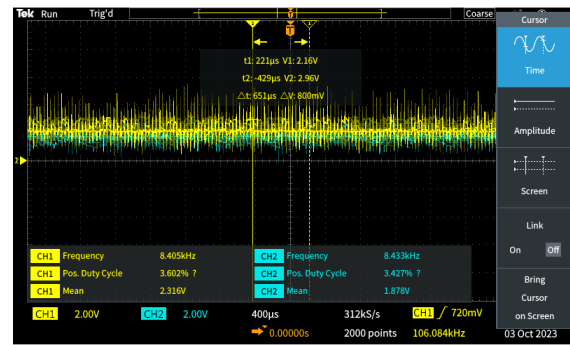
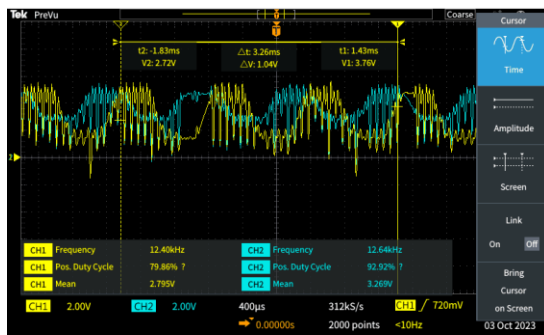
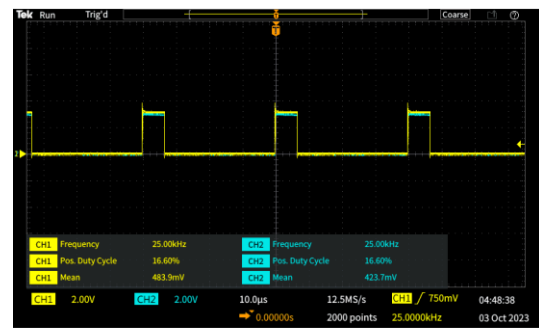
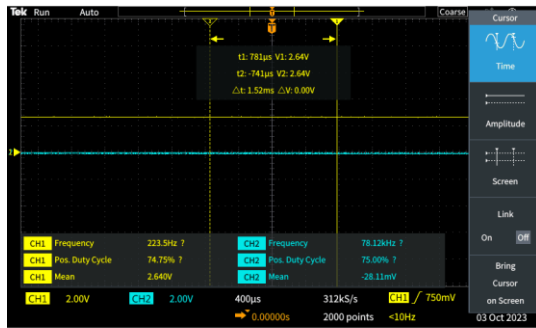


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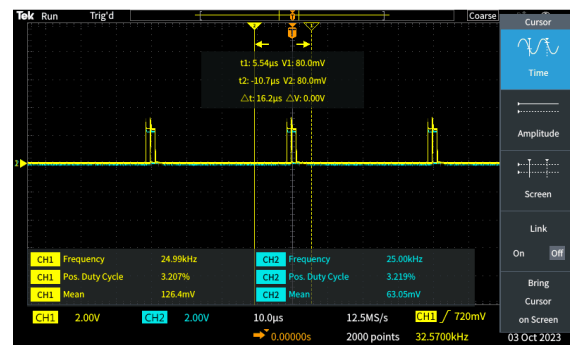
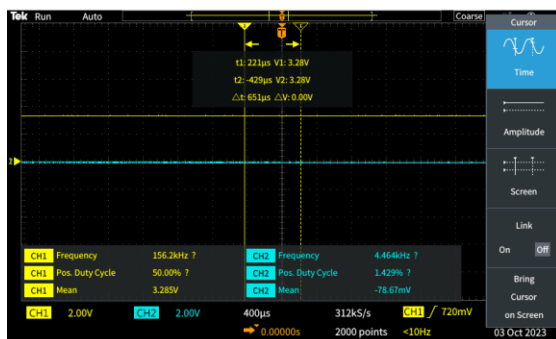


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- $2.64 V_{ref}$



- $3.28 V_{ref}$



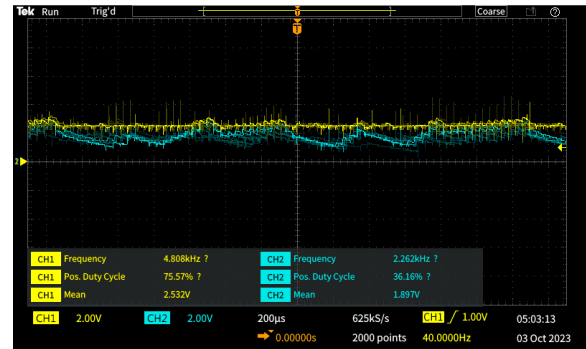
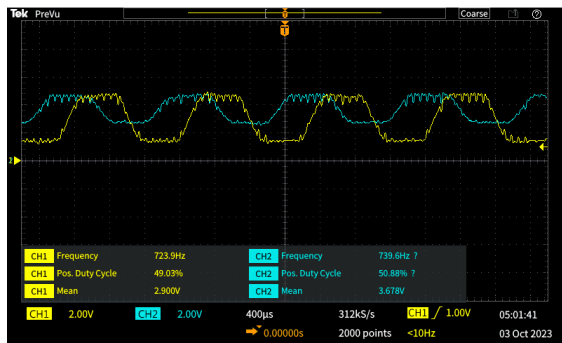
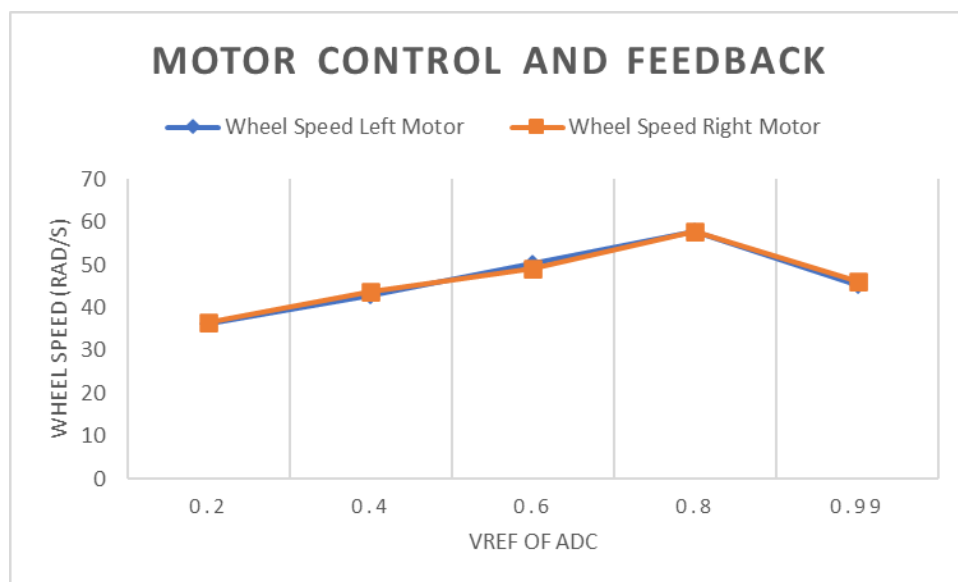


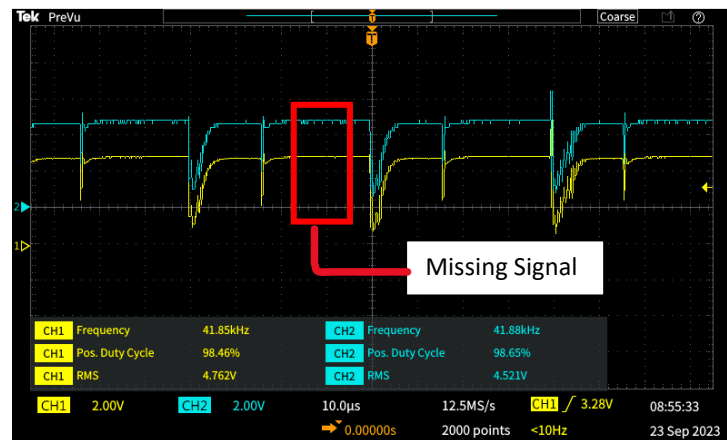
Table and Graph Results

Vref (V)	VrefA DC	Duty Cycle	Fencoder L (Hz)	Fencoder R (Hz)	Vout L (RMS)	Vout R (RMS)	VoutA DC	Wheel Speed L	Wheel Speed R	Motor Speed L	Motor Speed R
0.66	0.2	0.24	578	585	0.66	0.57	0.172727	36.125	36.5625	270.9375	274.21875
1.32	0.4	0.44	687	699	1.67	1.5	0.454545	42.9375	43.6875	322.03125	327.65625
1.99	0.6	0.632	806	787	1.95	1.79	0.542424	50.375	49.1875	377.8125	368.90625
2.64	0.8	0.83	926	926	2.32	1.88	0.569697	57.875	57.875	434.0625	434.0625
3.28	0.99	0.966	724	740	2.53	1.9	0.575758	45.25	46.25	339.375	346.875

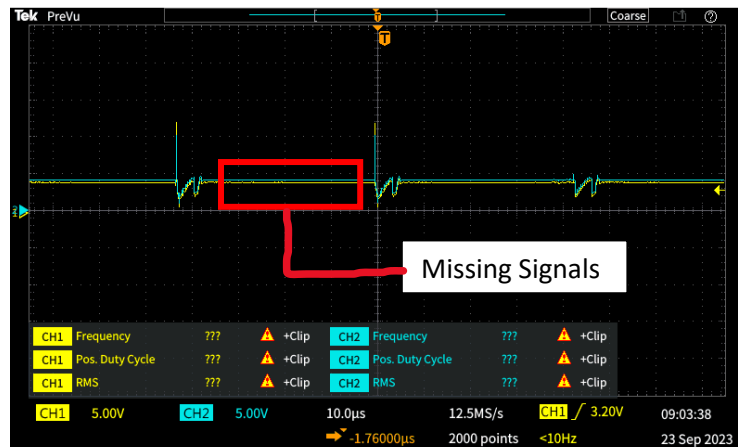


Problems Encountered

1. When switching from the OS-2 system of the micro controller to the OS-6, the printf function worked. However, the API functions of AnalogIn and PWMout did not work properly anymore.
2. Voltage decreased when the duty cycle increased. The low-side gate driver was inverting the input signal. So, the code was adjusted to account for this inversion.
3. The optical sensor was not able to properly produce a frequency relative to the wheel speed at high speeds. This may be due to a fault with the optical sensor or the slits in the wheel. As shown below, the sensor can only realize a few of the slits per one full-wheel rotation. Which in turn limited the output voltage from the voltage to frequency converter.



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TBS1052C - 9:04:14 PM 9/22/2023

Upon further inspection, the sensor was defective. We replaced the sensor and began to receive signals that were more accurate.

Questions

1. Why is the gate driver needed? VGS? VDS?

The gate driver is needed to boost the current supplied by the PWM of the micro-controller.

2. Why do we use a MOSFET? Why not a BJT?

MOSFETs are voltage-controlled and BJTs are current-controlled. We want a linear relationship between voltage and motor speed. So, we utilize a MOSFET.

3. In what mode(s) does a MOSFET act as a switch? An amplifier? How does this differ?

MOSFET's have three modes. Cut-off, triode and saturation. It will act as a switch in the cut-off and saturation region. It will act as an amplifier in the triode region.

4. Will the circuit in Figure 1 work without the diode? Explain.

The circuit in Figure 1 does not include a diode. It will work without a diode. However, a diode should be placed after the microcontroller to prevent damage to the microcontroller. We don't want the microcontroller to draw current from the circuit from the motor's internal capacitance.

5. Why do we use low-side configuration? (Hint: With a 7.2V Battery would the MOSFET turn completely off in high-side configuration? Justify.)

A low side configuration is used due to the low conduction losses given at higher duty cycles. This configuration includes analysis for power loss and has great efficiency with the gate driver.

Time sheet

Name	Date	Start Time	End Time	Total Time Spent	Tasks / Notes
Nicholas George	09/06/2023	20:00	21:30	1 Hr. 30 Min.	Created Lab Notebook and Lab 01 documentation
Nicholas George	09/09/2023	18:00	23:50	5 Hrs. 50 Min.	Lab 01 background
Team	09/11/2023	14:20	17:00	2 Hrs. 40 Min.	Circuit Design (Open-Loop Control)
Nicholas George	9/17/2023	13:00	18:30	5 Hrs. 30 Min	Circuit Design (Open-Loop Control) Circuit Design (Speed Measurement)
Team	9/18/2023	14:20	17:00	2 Hrs. 40 Min.	Circuit Design (Speed Measurement)
Team	9/19/2023	19:20	23:00	3 Hrs. 40 Min.	Circuit Design (Speed Measurement)

Team	9/22/2023	19:00	23:00	4 Hrs. 0 Min.	Circuit Design (Speed Measurement)
Nicholas George	9/24/2023	16:00	20:00	4 Hrs. 0 Min	Lab 01 Notebook, report, and Control Document

Lab 02: Model Identification

Motor Control (Group A)

General Objectives

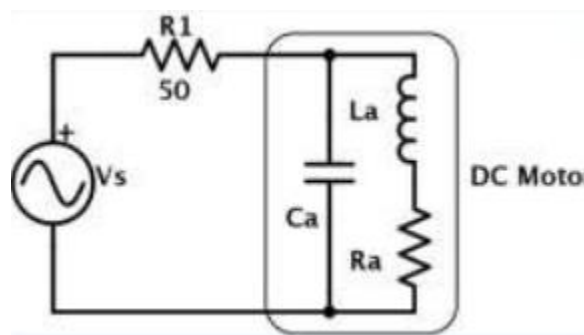
- Use programming techniques in mbed to solidify understanding of microcontroller features
- Design and implement a circuit used to perform the motor model identification process
- Validate system model using computer simulation

Main Specifications

- Motor voltage: 7.2V
- Switching frequency: ≥ 20 kHz
- Speed range: 0-100% (Design to match the DC motor's rated voltage!!!)
- External voltage (speed) reference: 0-3.3V
- Feedback: Hall Effect Sensor

Tasks: Part I

- Research data sheets for components and understand why they are used (their ratings)
- Measure the motors winding resistance.
 - Sample the resistance from several different points with respect to the shaft position and take the average.
 - When rotating the shaft, turn slowly, as to refrain from generating a voltage over the motor and inducing a current. The induced current could damage the Digital Multimeter.
 - Make sure to account for the leads of the DMM when measuring the resistance of the motor.
- Perform a frequency sweep, from one hertz to five megahertz using the function generator and the circuit shown.



- Tabularize the results with ten equally spaced points per decade (log scale).
- From the obtained data, determine the inductance 'L' of the motor.

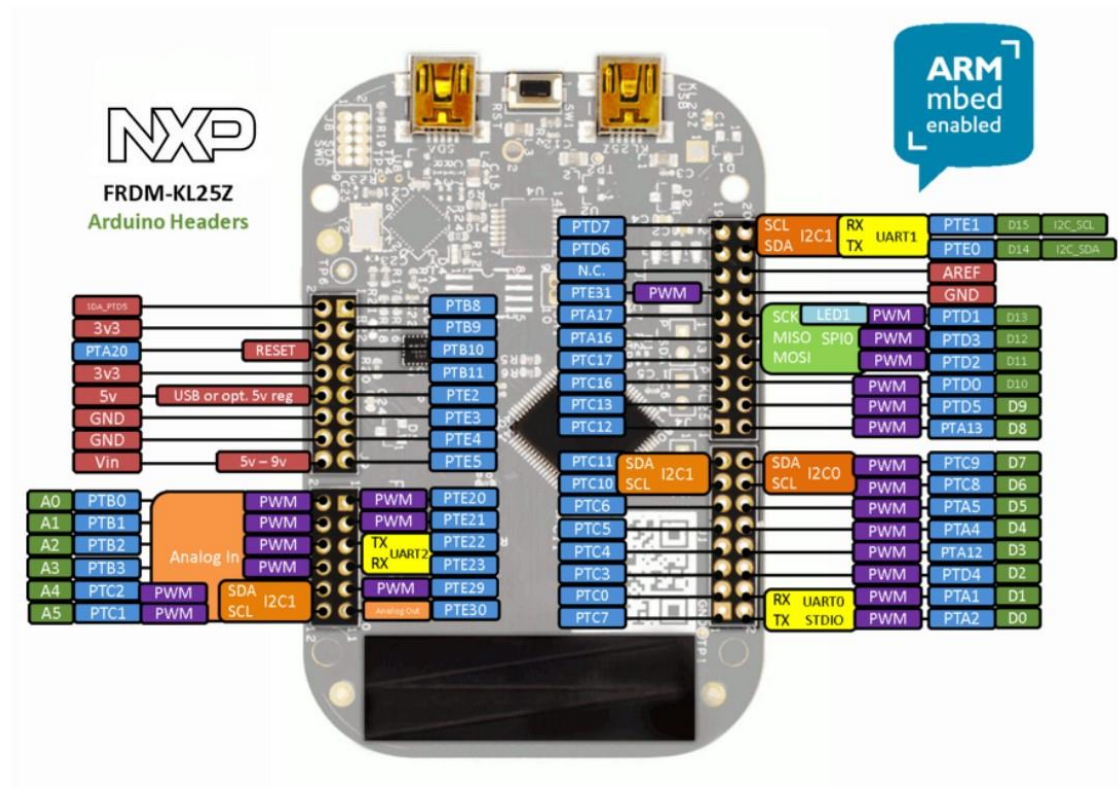
Tasks: Part II

- Derive the transfer function of a DC motor for angular speed
- Find 'KDC' from the data collected in laboratory exercise one.

-
- Find 'K' and 'B'
 -
 -
-
- Find ' τ_m ' and 'J'
 -
 -
 -
-
-
- Use the provided Simulink Model (Blackboard) to simulate the system
 - Explore the subsystem levels of the model and understand how these work
 - Add the appropriate source (Voltage) and sink (Speed) to run the simulation
 - Use a Matlab script to update the variables in the Matlab workspace.
 - After all, variables are solved and in the workspace run the simulation.
 - Perform a source voltage sweep from 0-100% similar to what you did in Laboratory exercise one and plot the average motor voltage vs. motor speed (rad/s).
 - Compare the simulated 'KDC' with the actual and calculate the percent error.
 - Determine the simulated ' τ_m ' and the corresponding percent error.

Background

Mbed Pinout



Mbed Datasheet

<https://www.nxp.com/docs/en/data-sheet/KL25P80M48SF0.pdf>

MOSFET (IRF540NPBF) Datasheet

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